

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
 B. Sc. Engineering 3rd Year 1st Term Backlog Examination, 2024
CE 3111
 (Structural Analysis and Design - I)

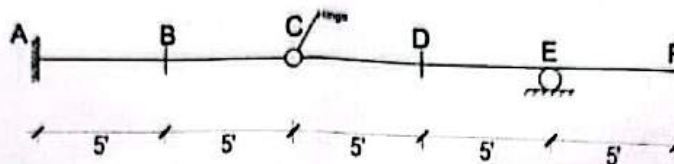
Full Marks: 210

Time: 3 hrs

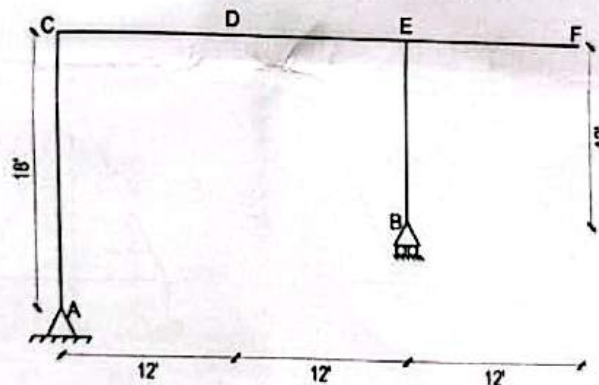
- N.B.** i) Answer any three questions from each section in separate script.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable value for any missing data.

Section - A

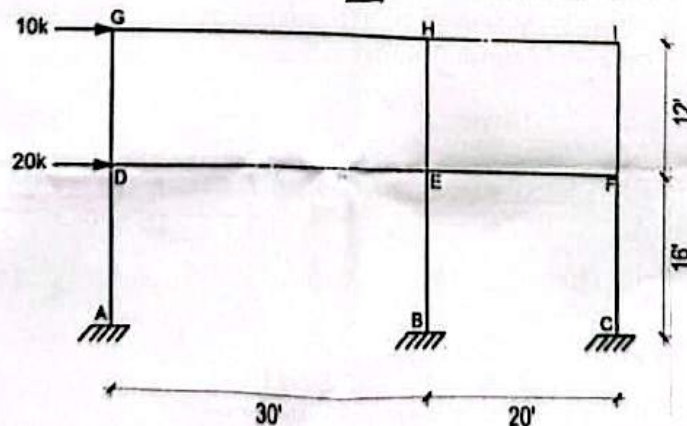
1. (a) Draw the influence lines for the vertical reactions at supports **A** and **E**, the reaction moment at support A, the shear at point B, and the bending moment at point D of the beam shown in figure. (18)



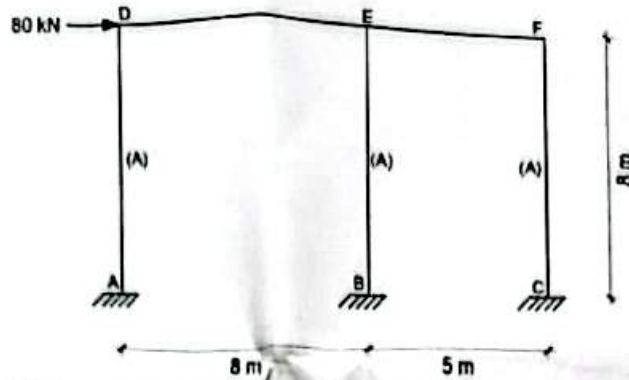
- (b) Draw the influence lines for the vertical reactions at supports A and B and the shear and bending moment at point D of the frame shown in figure. (17)



2. (a) Determine the approximate axial forces, shears and moments for all the members of the frame shown in figure by using the portal method. Also draw SFD and BMD. (17)

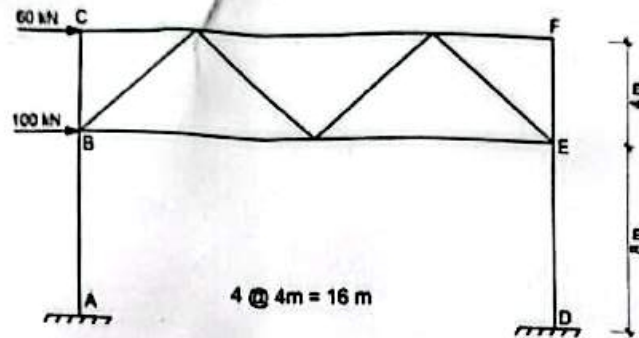


- (b) Determine the approximate axial forces, shears and moments for all the members of the frame shown in figure by using cantilever method. Also draw SFD and BMD. (18)



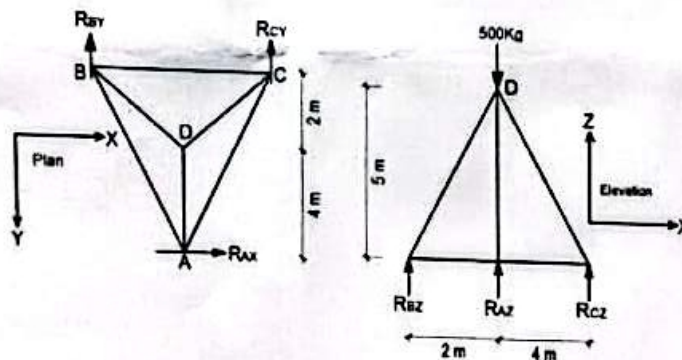
3. (a) Draw the SFD and BMD diagrams of columns AC and DF.

(15)



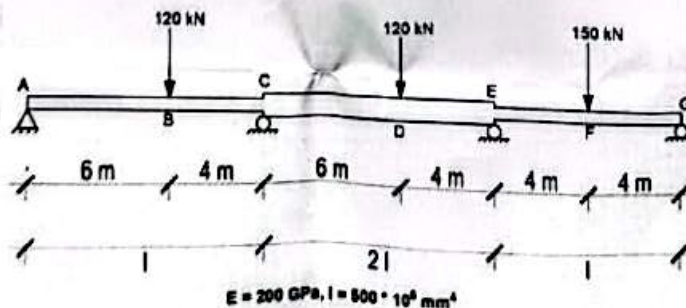
- (b) Determine the reactions and bar forces of Truss shown below.

(20)



4. (a) Determine the reactions and draw the shear and bending moment diagrams for the beam shown in figure using the three-moment equation, if the support settlements of 10 mm at A, 65 mm at C, 40 mm at E, and 25 mm at G.

(20)

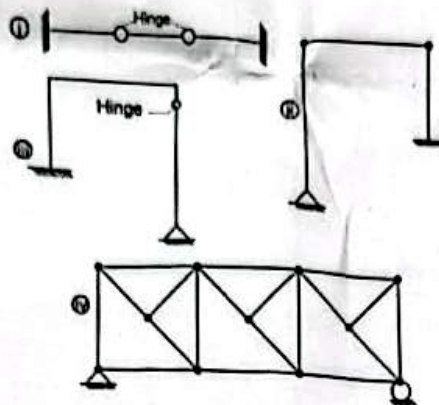


- (b) A suspension cable is supported at two points 20 m apart. The left support is 2 m above the right support. The cable is loaded with uniformly distributed load of 10

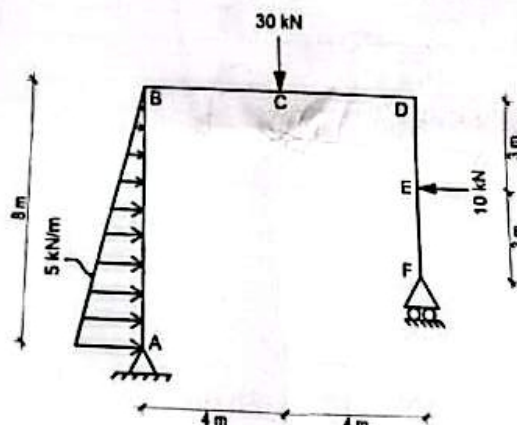
kN/m throughout the span. The maximum dip in the cable from left support level is 4 m. Find the maximum tension in the cable.

Section - B

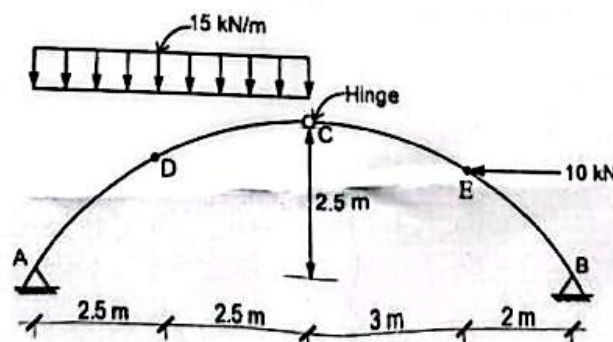
5. (a) Differentiate between: (i) Determinate and Indeterminate, (ii) Stable and unstable (06) structures with suitable examples.
 (b) Classify the structures shown in the figure below. Check the structures whether they (09) are stable or unstable and also statically determinate or indeterminate.



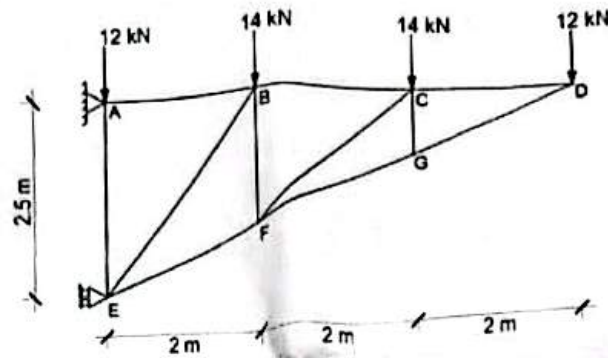
- (c) Determine the support reactions, and draw the axial force, shear force and bending (20) moment diagrams for the frame. Joints B and D are rigid joints.



6. (a) Determine the internal forces and moment at the point D in the three hinge parabolic (18) arch as shown in figure.



- (b) Compute the bar forces of the truss members FG, GD, CB and CF shown in following (17) figure.

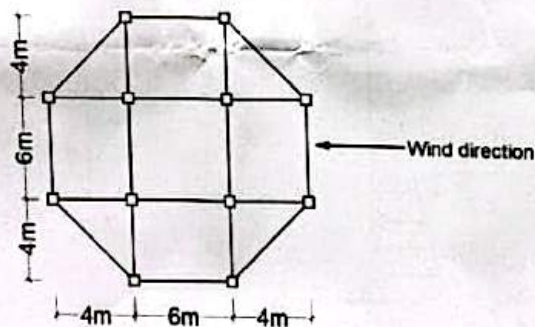


7. (a) Calculate the wind load at each floor level for a five storied building by using the following data- (18)

- Height of each floor = 3.2 m
- Structural importance co-efficient = 1.25
- Exposure category = B
- Basic wind = 190 Km/hr
- Pressure Co-efficient = 1.5

Table 7(a): Gust Response factors and combined height and exposure co-efficient

Height above GL (m)	G_H/G_z	C_z
0 - 4.5	1.32	0.80
6	1.29	0.87
9	1.26	0.97
12	1.23	1.06
15	1.22	1.13
18	1.20	1.19

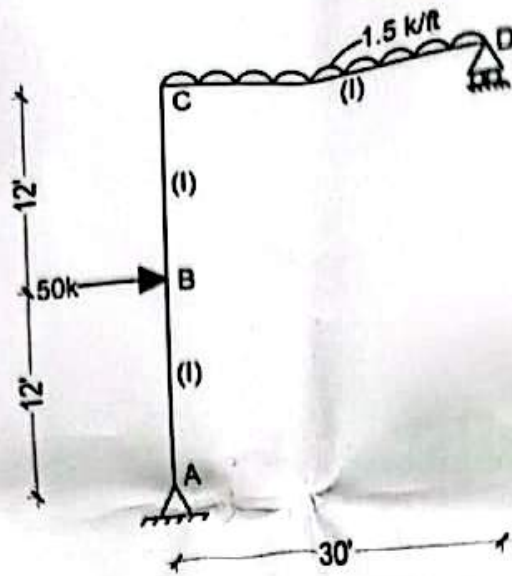


- (b) Using the equivalent static force method, calculate the earthquake load at each floor level for a five storey residential reinforced concrete building using the following data- (17)

- Height of each floor = 3 m
- Plan area = 20 m x 35 m
- Seismic zone co-efficient = 0.15
- Structural importance co-efficient = 1.25
- Response modification co-efficient = 8.0
- Seismic dead load = 2700 kN/ floor

Soil type	S	T_B (s)	T_C (s)	T_D (s)	η	C_t	m
SB	1.2	0.15	0.50	2.0	0.60	0.0466	0.90

8. (a) Determine the rotation and horizontal deflection of joint c of the frame shown in figure using unit load method. Given that, $E = 29000 \text{ ksi}$ and $I = 2500 \text{ in}^4$. (15)



- (b) Define conjugate beam. Write down the relationship between real and conjugate beam. Use the conjugate beam method to determine the slope and deflection at point B of the beam shown in figure. Assume, $E = 29,000$ Ksi and $I = 3000$ in⁴. (20)

